RIZVI COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science



PROJECT REPORT

Sentiment analysis

VADER, TextBlob and Hindi/Hinglish sentiment

Natural Language Processing (CSDO7011)

Submitted by

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NLP APPLICATION STUDY / 231A022

Sentiment analysis in context

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This study tests sentiment analysis for product feedback and social posts using two existing tools, VADER and TextBlob. It compares their actual local outputs on the same 36 project-authored inputs and implements a separate Hindi/Hinglish lexicon adapter. The original tools succeed on simple English cases but miss sarcastic complaints and Hindi sentiment. The adapter adds Hindi vocabulary, negation and intensity rules.

Application	Implemented study
Product and social-post sentiment	VADER + TextBlob; 36 unchanged test inputs
Indian language	Hindi/Hinglish dictionary, negation rules and an extension plan

How the experiment was run

The two tools receive the original text field, with no translation, transliteration, spelling correction, lowercasing or stopword removal. Only outer whitespace is stripped for custom inputs. VADER uses its distributed lexicon and rules; TextBlob uses its default English PatternAnalyzer. The original comparison uses the distributed package versions and declared fixed thresholds.

Positive means score >= +0.05, negative means score <= -0.05, and neutral falls between. The cutoffs follow VADER's documented recommendation. Applying the same cutoffs to TextBlob is an explicit project choice. VADER compound and TextBlob polarity have different scoring semantics, so their magnitudes are not directly calibrated. Subjectivity is not certainty; token sentiment proportions are not class probabilities.

Input design

The set contains six inputs in each of English, Negation, Sarcasm, Indian English, Hinglish and Hindi. Its reference distribution is 17 negative, 8 neutral and 11 positive. Examples and intended labels were prepared with AI assistance before inference, without independent annotation or native-speaker validation. The set deliberately stresses failure modes and is not representative of social-media traffic.

Scope follows the visible assignment image, which ends after the socio-linguistic challenges subpart. No unseen continuation was available.

NLP APPLICATION STUDY / 02

One input, two real tool responses

Requirement i: data supplied, observed response and success/failure analysis.

Both tools correctly label all six straightforward English inputs and all six negation examples under the declared references. For EN01, 'I love this phone. The camera is excellent!', VADER returns positive at +0.8478 and TextBlob returns positive at +0.7500. Preserving words such as 'not' allows both tools to label NG01, 'This phone is not good.', negative.

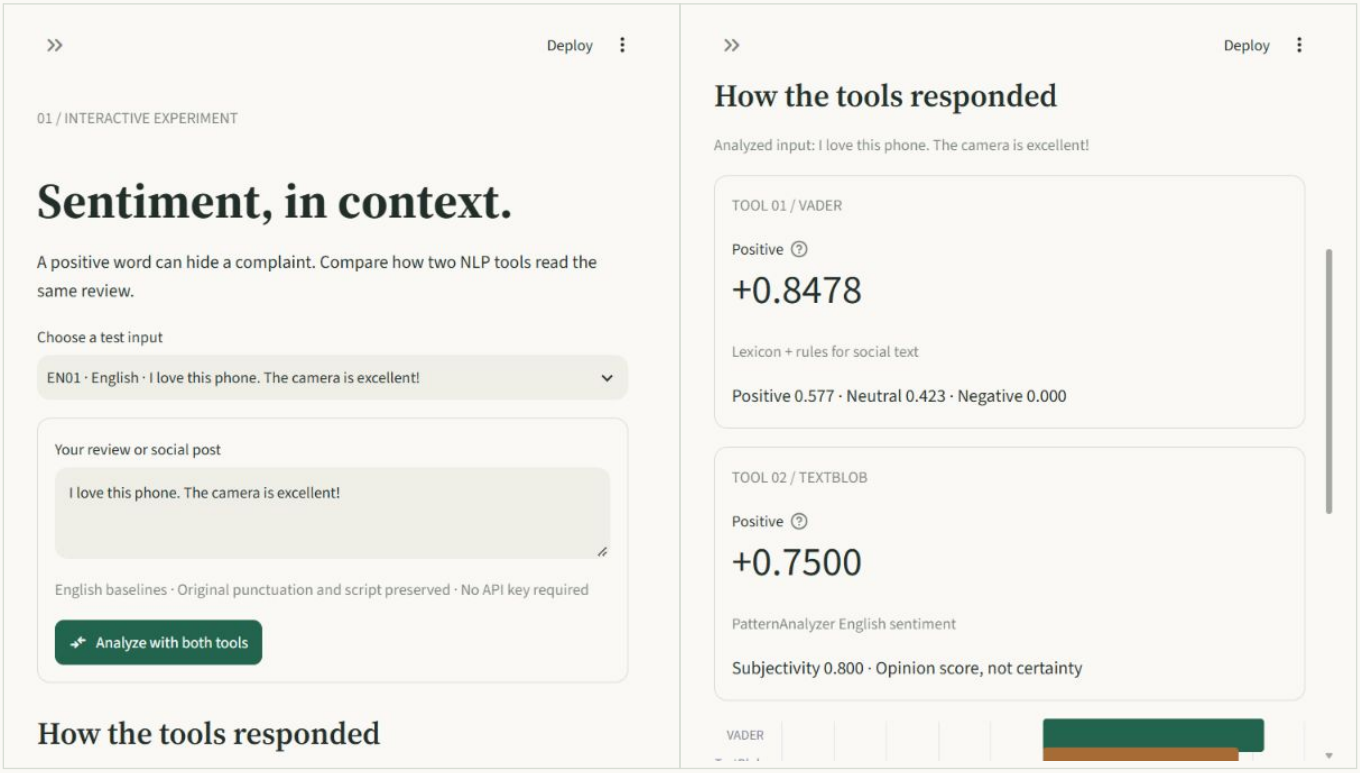


Figure 1. Actual app captures: EN01 entered in the interface, then both tool responses. The same sentence was passed to both packages.

Input ID	Tool response	Interpretation
EN01	VADER positive, +0.8478	Matches the positive reference.
EN01	TextBlob positive, +0.7500	Matches the positive reference.
NG01	Both negative for 'not good'	Negation is handled in this example.

A successful English example establishes that inference runs and that the tools can recognize explicit praise. It does not establish accuracy on sarcasm, unfamiliar idioms or another language.

NLP APPLICATION STUDY / 03

Measured comparison on 36 inputs

Requirement ii: same inputs in two tools and a comparison/error matrix.

Tool	Correct	Accuracy	Macro F1
VADER	18/36	50.0%	0.498
TextBlob	20/36	55.6%	0.555

TextBlob gets two more inputs correct on this set. Both exceed an always-negative baseline of 17/36, or 47.2%, by a small margin. Four output labels differ. This is a descriptive classroom experiment; the sample size and construction do not support a general ranking of the tools. Accuracy is correct labels divided by inputs. Macro F1 averages the three class F1 values, with undefined values set to zero.

Category	n	VADER accuracy	TextBlob accuracy
English	6	100.0%	100.0%
Hindi	6	33.3%	33.3%
Hinglish	6	33.3%	33.3%
Indian English	6	33.3%	66.7%
Negation	6	100.0%	100.0%
Sarcasm	6	0.0%	0.0%

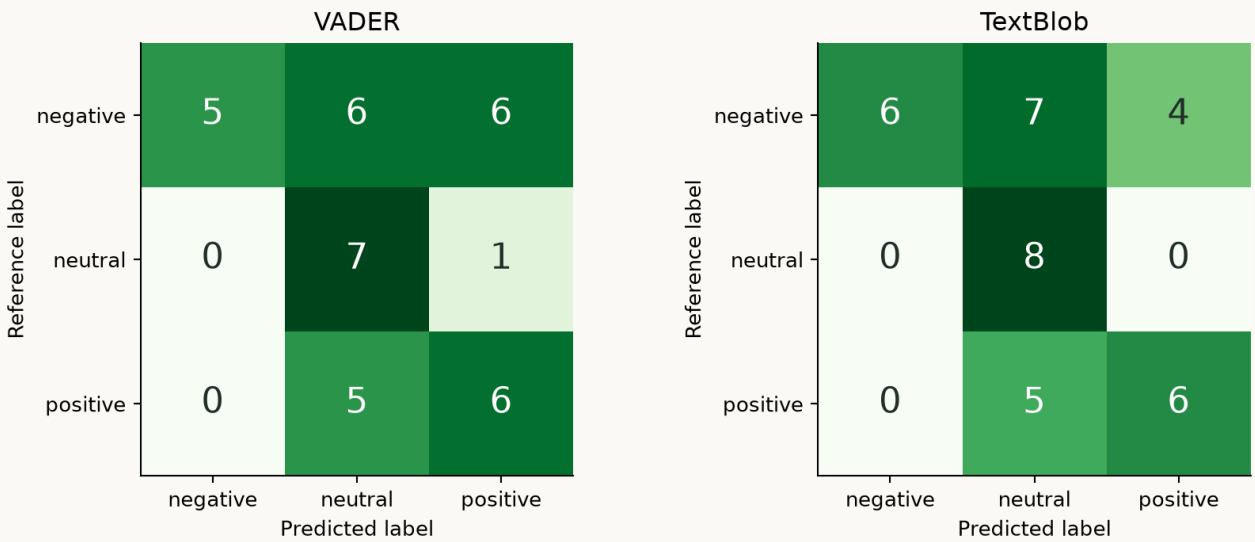


Figure 2. Confusion matrices from the actual predictions. Rows are references; columns are predictions. Class order is negative, neutral, positive. All 36 observations are counted for each tool.

Full-precision scores and all input IDs are preserved in results/comparison.csv and results/predictions.json.

NLP APPLICATION STUDY / 04

Output differences and error types

All four disagreements are shown below. A different output is not automatically a correct output.

ID and unchanged input	Reference	VADER / score	TextBlob / score
SA03 I just love paying for a broken product.	negative	positive +0.2732	neutral +0.0500
SA04 Fantastic! Another failed payment.	negative	positive +0.1511	neutral +0.0000
IN04 The restaurant is overhyped and the food is tasteless.	negative	neutral +0.0000	negative -0.6000
IN06 The shop opens after the Diwali holiday.	neutral	positive +0.4019	neutral +0.0000

Scores above are rounded. SA03's raw TextBlob polarity is 0.04999999999999999, just below 0.05. Full-precision classification is neutral although the display reads 0.0500. This disagreement includes a floating-point boundary effect.

Semantic and pragmatic errors

Both tools miss all six sarcastic complaints. SA01 says 'Great! The app crashed again and deleted my work.' The intended negative judgement conflicts with the surface praise. The observed outputs show a pragmatic/context error. They do not establish that either tool can infer the speaker's intent from isolated positive words.

SA03 and SA04 have positive VADER outputs and neutral TextBlob outputs; both are incorrect against their negative references. This is context loss rather than a measured translation error.

Cultural and regional expressions

Both tools return neutral for IN01, 'The food was simply paisa vasool.', although the intended reading is positive value for money. Both return neutral for IN03, 'The service is third class.', although the regional judgement is negative. IN02 includes 'I enjoyed it', and both tools succeed; that success alone does not demonstrate that they understood 'timepass'.

IN04 is a negative restaurant complaint. TextBlob labels it negative while VADER labels it neutral. IN06 is a factual Diwali schedule; TextBlob labels it neutral while VADER labels it positive. An English holiday-word association is a plausible explanation, but the output alone does not prove the cause.

No translation system or grammatical parser was evaluated, so the study does not claim measured translation or grammar errors. The evidence supports vocabulary, idiom, pragmatics and context failures.

NLP APPLICATION STUDY / 05

Hindi and Hinglish expose the gap

All six Hindi inputs and all six Hinglish inputs receive neutral from both tools. Four references in each group are positive or negative, so each tool gets only the two factual inputs right, or 2/6. Correct factual labels here do not show Hindi comprehension. DV01 means 'This phone is very good.' Both scores are zero despite the positive reference. HI04, 'Product accha nahi hai.', means the product is not good, yet both tools return neutral.

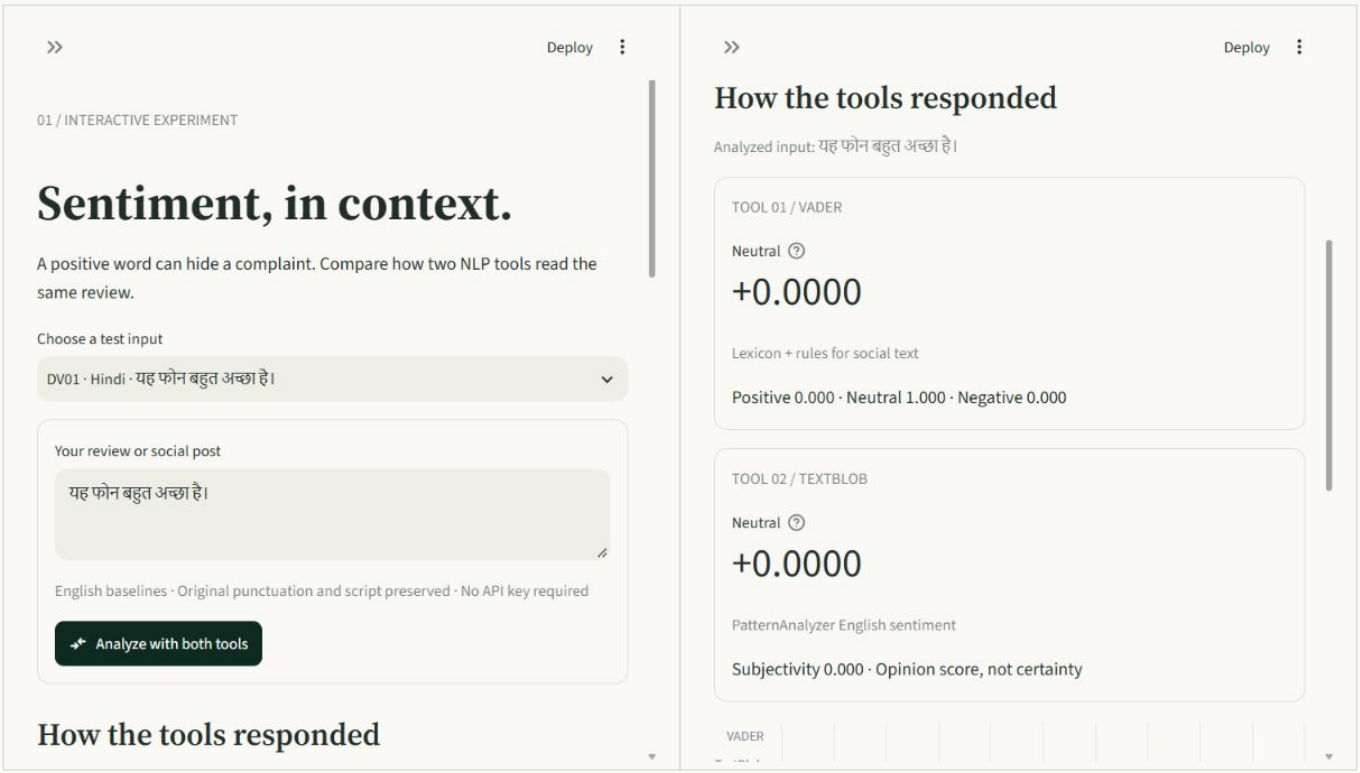


Figure 3. Actual DV01 input and observed neutral outputs. The native-script sentence is praise. Both tools return zero; agreement is not correctness.

Failure type	Concrete evidence
Script and vocabulary coverage	DV01 and DV03 express opposite sentiment in Hindi; both tools return neutral for each.
Romanization and code-switching	HI04, 'Product accha nahi hai.', includes Hindi negation over a product opinion. Both tools return neutral.
False appearance of success	The two factual inputs in each group are correctly neutral, despite identical zero outputs on the opinion inputs.

These observations motivate Hindi adaptation. The current app adds the Hindi adapter documented on the next page, while preserving these original tool measurements.

NLP APPLICATION STUDY / 06

Hindi sentiment in the updated app

Hindi and Hinglish inputs now use a project-authored sentiment dictionary with negation and intensity rules. The original English tools remain available in a collapsed comparison panel.

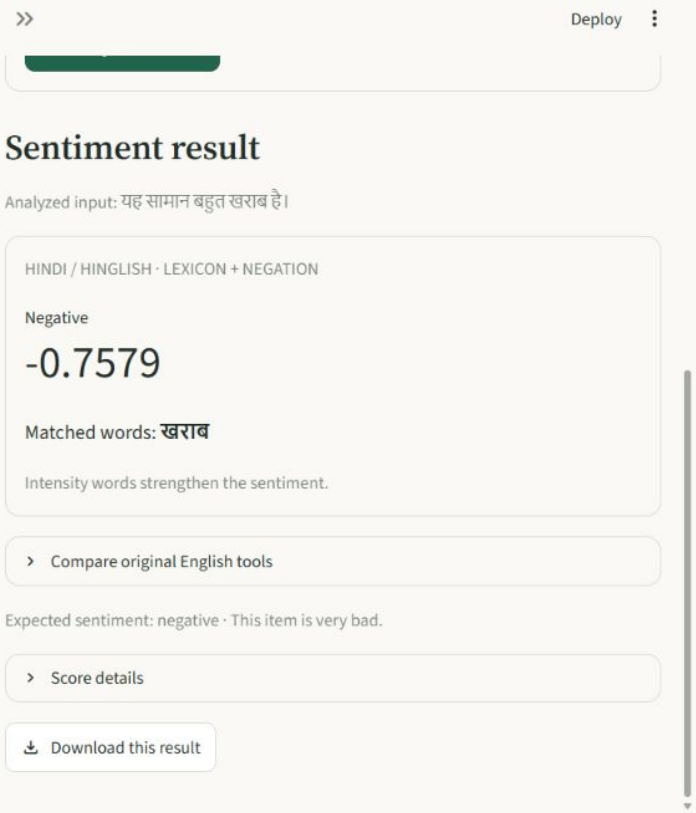


Figure 4. Current app response to the supplied Hindi complaint: negative at -0.7579. The matched term is kharab, strengthened by bahut.

The adapter sums matched word valences and normalizes the sum s as s / sqrt(s*s + 15). Nearby negation reverses a term with a 0.74 factor, while intensity words multiply its strength by 1.5. Hindi spelling variants and common Romanized forms share dictionary entries. Mixed-language English words use VADER's word lexicon.

All 12 coursework Hindi/Hinglish checks pass. Their vocabulary was available during development, so this is an illustrative check rather than an independent accuracy estimate. The full rules and measurements are stored in sentiment_lab/hindi.py and results/hindi_adapter.json.

NLP APPLICATION STUDY / 07

Extending the tool for Hindi

Requirement iii: linguistic resources and socio-linguistic challenges. The app implements a Hindi sentiment dictionary with negation and intensity rules; the steps below describe further extension.

Resource	Specific use
Labeled Hindi/Hinglish corpus	Collect a proposed 3,000-review pilot: 1,000 Devanagari, 1,000 Romanized Hindi and 1,000 mixed-language reviews.
Hindi sentiment lexicon	Native-speaker valence ratings and negation rules as an interpretable baseline.
IndicBERT and Hindi corpora	Fine-tune a pretrained encoder with sentiment labels. Pretraining alone is not sentiment classification.
Unicode tokenization and language ID	Preserve Hindi/English tokens, emoji and negation. Optional Hindi POS/dependency tagging helps inspect scope errors.
IndicXlit; optional IndicTrans2	Compare transliteration and translation as separate pipelines while retaining original input and checking meaning loss.

Implementation and evaluation steps

1. Collect consented reviews across products, delivery, food and social posts. Remove identifying information and group duplicates or shared conversations.

2. Have two Hindi speakers independently label each example. A third adjudicates disagreements. Record sarcasm, mixed aspects and regional phrases; measure agreement before adjudication.

3. Normalize Unicode to NFC while retaining negation, emoji, punctuation and repeated letters. Store original and normalized text. Cover accha/achha/acha and nahi/nahin variants.

4. Split grouped data into 70% training, 15% validation and 15% test with script/domain balance. Fine-tune using training data and select settings using validation only. Keep the 36 demonstration examples out of training.

5. Compare a Hindi lexicon baseline and fine-tuned encoder on the same untouched test IDs. Report macro F1, per-class recall, confusion matrices, subgroup sizes and confidence intervals by script, dialect, domain and code-switching level.

Regional Hindi varieties, politeness, sarcasm and code-switching need speaker/context coverage. Avoid treating Latin script as proof of English. Mixed-aspect reviews may need aspect labels or human review. Add abstention only after validation; no improvement numbers are claimed.

NLP APPLICATION STUDY / 08

Limits, evidence and reproduction

NG04, 'I do not hate this movie.', has a weak-positive reference, but a neutral reading is defensible. NG05 and NG06 contain mixed aspects and use the final product-related clause for the overall label. Sarcasm references are all negative. These annotation choices limit interpretation. The results were not used to revise labels or remove examples.

The app uses synthetic examples rather than scraped Twitter or Facebook data. It combines the original two-tool study with an implemented Hindi sentiment dictionary and negation rules. The Hindi checks use illustrative vocabulary available during development; broader validation requires a new corpus.

Reproduce the project

Use Python 3.12 in the NLP_Assignment-1/Sentiment analysis directory. Install requirements-dev.txt, then run these commands:

```
python -m scripts.evaluate

python -m scripts.evaluate_hindi

python -m pytest -q

python -m scripts.build_notebook

python -m scripts.build_report

python -m streamlit run app.py
```

Verified runtime: Python 3.12.14, VADER 3.3.2, TextBlob 0.20.1. The provenance JSON records the UTC run time, package versions, threshold and dataset SHA-256.

Submission evidence

The executed notebook contains actual tool outputs. The CSV contains all 36 paired predictions, reference labels and challenge notes. The app supplies input/output JSON downloads, filters and error matrices. Actual screenshots are stored in reports/screenshots. The full Hindi resource and socio-linguistic plan is in docs/hindi_adaptation.md.

Implementation, illustrative examples and documentation were prepared with AI assistance. Reported measurements were produced by the named tools; screenshots came from the running application.

Sources consulted 24 October 2026

VADER implementation, paper and scoring
https://github.com/cjhutto/vaderSentiment

TextBlob sentiment and API documentation
https://textblob.readthedocs.io/en/stable/quickstart.html#sentiment-analysis

AI4Bharat IndicBERT
https://indicnlp.ai4bharat.org/pages/indic-bert/

AI4Bharat IndicXlit
https://github.com/AI4Bharat/IndicXlit

AI4Bharat IndicTrans2
https://github.com/AI4Bharat/IndicTrans2